NETFLOW

LIQUID FLOW MEASUREMENT · NT-VMF SERIES

ELECTROMAGNETIC FLOW METER

Precise flow measurement for conductive liquids with no moving parts and zero maintenance

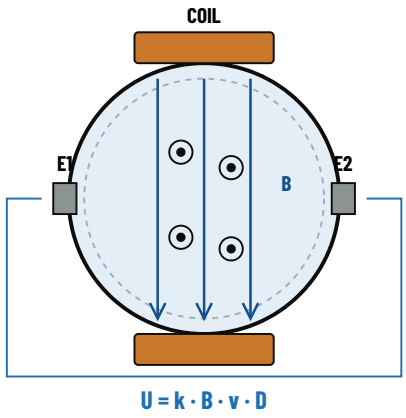


The NT-VMF series electromagnetic flow meters operate based on Faraday's law of induction. Since there is no constriction or moving parts in the measuring tube, there is no pressure drop or wear. With a variety of liner and electrode materials, they are suitable for a wide range of conductive liquids—from clean water to wastewater, acid and alkali solutions to food products.

DN3 – DN3000	±%0,5	0,2 – 15 m/s	IP65 / IP68
NOMINAL DIAMETER	ACCURACY · OPT. ±0.2%	FLOW VELOCITY	PROTECTION CLASS

WORKING PRINCIPLE

FARADAY'S LAW OF INDUCTION



1 · MAGNETIC FIELD

The coils on the measuring tube generate a constant magnetic field (B) perpendicular to the flow direction.

2 · INDUCED VOLTAGE

As the conductive liquid passes through the field, a voltage (U) proportional to the flow velocity (v) is induced. This voltage is read by the electrodes (E1, E2) on both sides of the tube.

3 · FLOW CALCULATION

The converter calculates the instantaneous flow rate from the measured voltage and internal diameter (D); records forward, reverse and net totals.

FEATURES

- Measurement independent of pressure, temperature, density and viscosity
- Bidirectional flow measurement, three built-in totalizers
- Self-diagnostics, empty pipe and limit alarms
- RS485 Modbus RTU standard · HART, GPRS, Profibus optional
- No moving parts, no pressure loss, maintenance-free
- Built-in grounding electrodes
- Compact or remote type, standard cable 10 m, max. 100 m
- Power failure logging function (optional)

APPLICATION AREAS

Drinking water · Raw water · Wastewater · Seawater · Cooling water · Heat exchangers · Brine solutions · Acid and alkali solutions · Pulp and beverages · Industrial processes

TECHNICAL SPECIFICATIONS

MEASUREMENT

Nominal diameter	DN3 – DN3000 (⅛" – 120")
Accuracy	±0.5% of reading · opt. ±0.2% (≥0.5 m/s)
Repeatability	≤ %0,17
Flow velocity	0,2 – 15 m/s
Min. conductivity	5 µS/cm · demineralized water >20 µS/cm
Fluid temperature	Compact –20 ... 80 °C · remote –20 ... 120 °C
Ambient temperature	–25 ... 60 °C

SENSOR

Measuring tube	SS304
Flange and body	Carbon steel · SS304 / SS316 optional
Elektrot	SS316L · Hastelloy B / C · titanium · tantalum · platinum
Flange standard	DIN · ANSI · JIS
Nominal pressure	PN10 / 16 / 25 / 40 · 10K / 20K / 30K · 150# / 300# / 600#
Protection class	IP65 / IP67 · IP68 for remote type (submersible)
Bobin	%99 copper
Kalibrasyon	TÜRKAK AB-0354-K · ±%0,27
Belgeler	CE · ISO 9001

CONVERTER

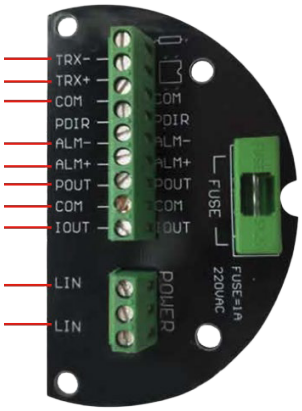
Power supply	85 – 250 VAC (50/60 Hz) · 20 – 36 VDC
Power consumption	< 20 W
Analog output	4 – 20 mA (load 0 – 750 Ω)
Frequency output	Forward / reverse flow · 1 – 5000 Hz
Alarm output	2 × isolated open collector (OCT)
Communication	RS485 Modbus RTU · ops. HART, GPRS, Profibus
Display	Backlit LCD · 3 lines · 4 keys
Displayed values	Instantaneous flow · total flow · flow velocity · flow percentage · empty pipe ratio
Birimler	L/s · L/m · L/h · m^3/s · m^3/m · m^3/h · gal · kg · t
Housing	Aluminum alloy, epoxy coated
Protection class	IP65 · ops. IP67
Dil	Turkish · English

LINER AND ELECTRODE SELECTION

TOP ROW LINER · BOTTOM ROW ELECTRODE

PTFE –40 ... 180 °C Acids, high temperature	**HARD RUBBER** –10 ... 80 °C Clean and raw water	**NEOPREN** –10 ... 70 °C Wastewater, low pressure	**F46 (FEP)** –20 ... 120 °C Food, medium temperature chemical	**PFA** –40 ... 180 °C High purity, pharmaceutical	**POLYURETHANE** –10 ... 60 °C Abrasive slurry, mining
SS316L Water, wastewater, hafif kimyasallar	**HASTELLOY B** HCl, fosforik, oksitleyici olmayan asitler	**HASTELLOY C** HNO_3, H_2SO_4, chlorinated, oxidizing agents	**TITANIUM** Seawater, tuzlu su, hipoklorit	**TANTAL** Kral suyu, hemen all chemicals	**PLATINUM** Highly corrosive, special applications

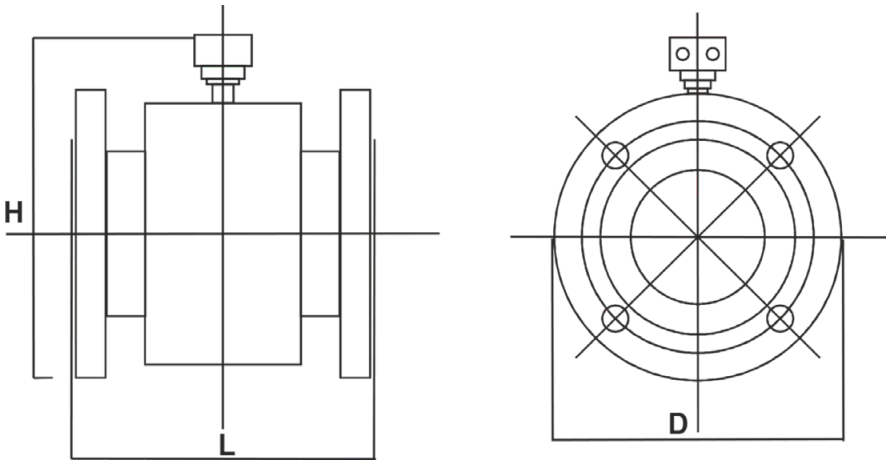
CONVERTER CONNECTION BOARD



TRX– / TRX+	RS485 Modbus RTU
COM	Communication ground
PDIR	Flow direction output
ALM–	Low limit alarm
ALM+	High limit alarm

POUT	Frequency / pulse output (bidirectional)
COM	Output ground
IOUT	4 – 20 mA current output
L / N	Power supply · sigmedium 1 A

DIMENSIONS FLANGED TYPE · mm



DN	PN (MPa)	L · PTFE	L · NEOPREN	D	H	WEIGHT (kg)
10	4,0	200	–	90	220	6
15	4,0	200	–	95	220	8
20	4,0	200	–	105	220	10
25	4,0	200	–	115	223	12
32	4,0	200	–	140	240	13
40	4,0	200	200	150	250	14
50	4,0	200	200	165	263	15

DN	PN (MPa)	L · PTFE	L · NEOPREN	D	H	WEIGHT (kg)
65	4,0	200	200	185	283	18
80	4,0	200	200	200	290	20
100	4,0	250	250	235	318	25
125	4,0	250	250	270	350	28
150	4,0	300	300	300	380	30
200	1,6	350	350	340	430	50
250	1,6	450	450	405	495	70
300	1,6	500	500	460	547	95

MEASUREMENT RANGE FLOW RATE m^3/h · BY FLOW VELOCITY

DN	0,1 m/s	0,5 m/s	1 m/s	4 m/s	10 m/s	15 m/s
15	0,064	0,318	0,636	2,543	6,359	9,538
25	0,177	0,883	1,766	7,065	17,66	26,49
50	0,707	3,533	7,065	28,26	70,65	106,0
80	1,81	9,04	18,09	72,35	180,9	271,3
100	2,83	14,13	28,26	113,0	282,6	423,9
150	6,36	31,79	63,59	254,3	635,9	953,8
200	11,3	56,52	113,0	452,2	1130	1696
300	25,43	127,2	254,3	1017	2543	3815
500	71	353,3	707	2826	7065	10598
1000	283	1413	2826	11304	28260	42390

Recommended operating velocity 0.5 – 4 m/s. Complete table for all sizes from DN3 to DN3000 available upon request.

COMPACT AND REMOTE TYPE



COMPACT TYPE

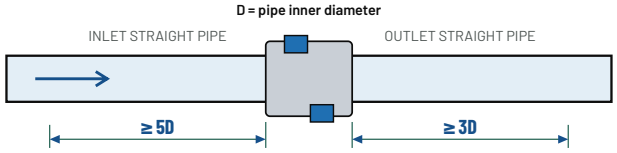
Converter is mounted on the sensor. Fluid temperature –20 ... 80 °C. Quick installation, single-piece solution.



REMOTE TYPE

Converter is mounted separately from the sensor; cable standard 10 m, max. 100 m. Fluid temperature –20 ... 120 °C, sensor can be used underwater with IP68.

INSTALLATION KURALLARI



- The measuring tube must always be completely full; do not mount at the highest point of the pipe or at a free-discharge outlet.
- In vertical lines, the flow direction must be from bottom to top.
- Install on the discharge side of the pump, not the suction side.
- Place control valves on the outlet side of the flow meter.
- The electrode axis must be horizontal; a vertical electrode axis is affected by air bubbles and sediment.
- Keep away from strong magnetic fields, large motors and frequency inverters.

TOPRAKLAMA

- On metal pipes, the flow meter is connected to the pipe and ground via the flanges.
- On plastic or internally lined pipes, grounding rings are used on both flanges.
- Grounding is done to the PE terminal with a copper conductor having a cross-section larger than 1.6 mm^2 and resistance lower than 10 Ω.

ORDER INFORMATION

To select the correct model, simply provide the following information:

- Fluid and conductivity
- Min. / maks. debi
- Pressure class, flange standard
- Compact / remote type
- Nominal diameter (DN)
- Process temperature
- Astar ve elektrot
- Power supply, output, communication

TÜRKAK
ISO/IEC 17025:2017

ACCREDITATION	UNCERTAINTY	SCOPE	RECOGNITION
AB-0354-K	**±%0,27**	**1 – 350 m^3/h**	**EA · ILAC MRA**

Every flow meter is calibrated in our TÜRKAK accredited calibration laboratory before shipment; a traceable calibration certificate is shipped with the device.

Contact our technical team for diameter, liner, electrode and pressure class selection.

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